Waypoint RIT — 3D Smart Campus Blueprint

Next-Generation Interactive 3D Spatial Navigation & Indoor GIS Ecosystem

Project Title: Waypoint RIT Campus GIS	**Event Category:** Innovation Showcase / Smart Campus
Technology Stack: FastAPI, Mapbox GL JS WebGL, MySQL, GeoJSON	**Status:** Fully Integrated Multi-Platform Release

1. Executive Summary & Project Vision

Traditional campus maps are static 2D images that fail to communicate complex multi-story building layouts, room locations, or true indoor spatial awareness. **Waypoint RIT** solves this challenge by introducing a state-of-the-art **3D Interactive GIS System** designed specifically for large academic institutions.

By synthesizing real-time GPS localization, high-precision architectural room buffering, and WebGL 3D extruded rendering, Waypoint RIT empowers students, faculty, visitors, and event attendees to explore campus buildings, isolate specific floor cutaways, and navigate turn-by-turn both outdoors and indoors.

Architectural X-Ray Inspection	Robust Data Resiliency	Unified Ecosystem
Interactive 3D building shells slice dynamically by floor, revealing 0.38m thick solid walls and 3D room tags.	Auto-switching local/cloud APIs paired with zero-crash database fallback ensure flawless event demos.	Synchronized desktop kiosk, interactive admin dashboard, and touch-first mobile interfaces.

2. 3D Architectural GIS & Rendering Engine

Waypoint RIT employs Mapbox GL JS coupled with custom client-side GeoJSON computational geometry to construct a fully volumetric digital twin of the RIT campus:

- **Extruded Building Shells (`building-shells`):** Renders 3D building blocks with dynamic hover styling and automatic X-Ray cutaway filtering upon user selection.
- **Elevated Floor Plates (`indoor-floor-plate`):** Dynamically positions floor slab polygons at precise `elevation = level * 4.0m` heights.
- **0.38m Thick Solid Architectural Walls (`indoor-walls`):** Generates real-world 38cm perimeter wall buffers around every room footprint, extruded to 2.6m floor-to-ceiling height to prevent sub-pixel WebGL clipping.
- **Mercator 3D Projected Labels (`roomMarkersList`):** Uses high-precision `transform.projMatrix` calculations to lock HTML room tags directly over elevated 3D room centers.

3. Unified Multi-Platform Frontend Ecosystem

To cater to diverse campus use cases—from interactive campus kiosks to on-the-go smartphone navigation—Waypoint RIT provides three tailored interfaces:

Interface Layer	Target Port	Core Functionality & UX Specialization

Main Kiosk (`frontend`)	Port 8080	Public-facing 3D campus explorer with interactive info panel, floor selector widget, and high-visibility indoor turn-by-turn routing.
Interactive New / Admin (`frontend_new`)	Port 8082	Advanced responsive desktop/kiosk interface with full room directory search, live campus announcements, and X-Ray floor slicing.
Mobile Explorer (`frontend_mobile`)	Port 8081	Touch-optimized mobile web app featuring spring-animated bottom sheets, GPS Locate-Me tracking, and instant room selection.

4. Backend API & Zero-Crash Resiliency Architecture

The server-side engine is built on **FastAPI (Python)** serving structured JSON and GeoJSON geometry endpoints (`/admin/buildings`, `/admin/rooms`, `/live-data`, `/search/{query}`).

Zero-Crash Event Fallback Mechanism: For high-stakes presentations and technical showcases, the API layer incorporates automated fallback interception. If local MySQL connectivity is interrupted or unavailable, the backend immediately transitions to in-memory campus geometry polygons, guaranteeing 100% uptime without manual intervention.

Dynamic Host Resolution Engine: All frontends auto-detect their network origin (`isLocal`), routing seamlessly to `127.0.0.1:8000` during local showcase execution and `waypoint-rit.onrender.com` in production deployment.

5. Event Showcase Demonstration Workflow

Step #	Demonstration Action	Observed System Behavior & Innovation Highlight
01	Campus Overview Fly-In	WebGL camera glides smoothly to RIT Campus center at 60° isometric pitch with 3D atmospheric lighting.
02	Building X-Ray Selection	Clicking any academic block (e.g., C Block) hides the outer shell and reveals elevated floors and room cards.
03	Floor-by-Floor Slicing	Tapping 'Floor 7' elevates 38cm solid architectural walls and projects WebGL room tags locked over classrooms.
04	Clean Dismiss / Reset	Unified `deselectBuilding()` restores full campus shells and returns camera to global campus view instantly.